

CCSU BCA Semester 2 – Computer Organization

2026 Exam-Oriented Important Topics + Short Notes

1. Basic Organization

Short Note: Computer organization describes how hardware units are connected and operate. Learn CPU, memory, I/O and system bus concepts.

Exam focus: Definition + key points + example/diagram/program where applicable.

2. CPU & Registers

Short Note: CPU includes ALU, control unit and registers. Learn PC, IR, accumulator and general-purpose registers.

Exam focus: Definition + key points + example/diagram/program where applicable.

3. Instruction Cycle

Short Note: Fetch the instruction from memory, decode it, execute it and store/update the result. Understand the role of clock/control signals.

Exam focus: Definition + key points + example/diagram/program where applicable.

4. Memory Hierarchy

Short Note: Registers are fastest and smallest, followed by cache, main memory and secondary storage. The hierarchy balances speed, cost and capacity.

Exam focus: Definition + key points + example/diagram/program where applicable.

5. Cache Memory

Short Note: Cache is small, fast memory placed close to CPU to reduce average memory-access time. Learn locality and basic mapping concepts.

Exam focus: Definition + key points + example/diagram/program where applicable.

6. I/O & Interrupts

Short Note: I/O transfers data between computer and peripherals. Interrupts allow a device/event to request CPU attention. Learn polling vs interrupt-driven I/O.

Exam focus: Definition + key points + example/diagram/program where applicable.

7. DMA

Short Note: Direct Memory Access permits data transfer between an I/O device and memory with reduced CPU involvement. Learn controller, transfer and completion concepts.

Exam focus: Definition + key points + example/diagram/program where applicable.

■ Important Questions

- Explain **Basic Organization** with suitable example/diagram/formula/program.
- Explain **CPU & Registers** with suitable example/diagram/formula/program.
- Explain **Instruction Cycle** with suitable example/diagram/formula/program.
- Explain **Memory Hierarchy** with suitable example/diagram/formula/program.
- Explain **Cache Memory** with suitable example/diagram/formula/program.
- Explain **I/O & Interrupts** with suitable example/diagram/formula/program.
- Explain **DMA** with suitable example/diagram/formula/program.

Note: These are revision notes aligned to the CCSU BCA curriculum structure available for 2025–26. Check your college's paper combination/electives before the exam.